IGCSE 0607 Practice Questions — E9.1 — Combined Questions & Markscheme

Introduction to probability

Total Questions: 6 — **Total Marks:** 32 Marks

Table of Contents & Quick Navigation

- **E9.1** Introduction to probability
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(2 Qs)*

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1 A biased spinner can land on 1, 2, 3, 4, or 5.
The table shows the probabilities of the spinner landing on each of the numbers.

Outcome	1	2	3	4	5
Probability	0.15	$2x$	0.31	x	0.24

(a) Find the value of x.

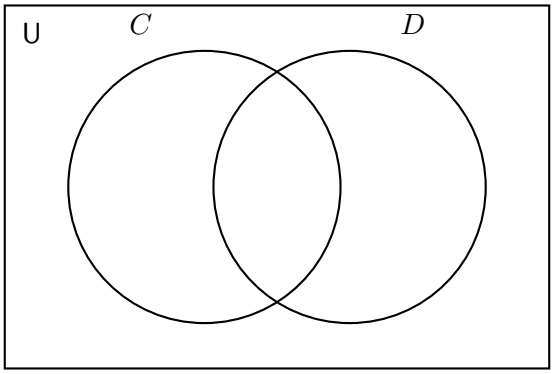
$x =$ ..... **[2]**

(b) The spinner is spun once.
Find the probability that it does not land on 2.

..... **[2]**

2 A group of 120 people were asked about the pets they own.
$C = \{\text{people who own a cat}\}$
$D = \{\text{people who own a dog}\}$
$n(\mathsf{U}) = 120$
$n(C) = 55$
$n(D) = 68$
18 people own neither a cat nor a dog.

(a) Complete the Venn diagram to show the number of people in each region.



[3]

(b) A person is chosen at random.
Find $P(C \cap D)$.

..... **[1]**

(c) Find $P(D')$.

..... **[2]**

(d) Find the probability that a person chosen at random owns a cat or a dog, but not both.

..... **[2]**

3 A bag contains only red counters, blue counters, and green counters.
The number of red counters is x^2.
The number of blue counters is $3x$.
The number of green counters is 2.

(a) Write down an expression, in its simplest form, for the total number of counters in the bag.

..... **[1]**

(b) A counter is chosen at random.
The probability that it is red is $\frac{8}{15}$.
Show that $7x^2 - 24x - 16 = 0$.

[2]

(c) Solve the equation $7x^2 - 24x - 16 = 0$ to find the value of x.

$x =$ ..... **[3]**

(d) Find the probability that a randomly chosen counter is not blue.
Give your answer as a fraction in its simplest form.

..... **[2]**

4 Event A is that a train arrives on time.

$P(A) = 0.83$

(a) Work out $P(A')$.

..... **[1]**

(b) State in words what the event A' represents.

..... **[1]**

5 A biased spinner can land on 1, 2, 3 or 4.
The table shows the probabilities of the spinner landing on each number.

Number	1	2	3	4
Probability	$\frac{1}{5}$	0.35	x	$2x$

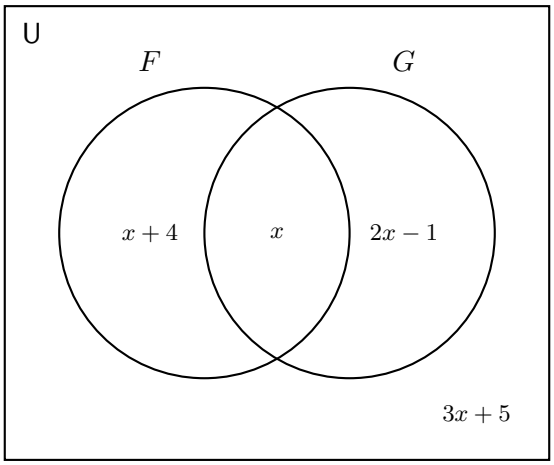
(a) Work out the value of x.

$x =$ ..... **[2]**

(b) Find the probability that the spinner does not land on 4.

..... **[2]**

6 The Venn diagram shows information about the number of students in a class of 50 who study French (F) and German (G).



(a) Work out the value of x.

$x =$ ..... **[2]**

(b) A student is chosen at random.
Find $P(F')$.

..... **[2]**

(c) A student is chosen at random.
Find the probability that this student studies exactly one of these languages.

..... **[2]**

1 **(a)** $0.15 + 2x + 0.31 + x + 0.24 = 1$ or $3x + 0.7 = 1$ **[M1]**

0.1 **[A1]**

(b) $P(2) = 2(0.1) = 0.2$ **[M1]**

0.8 **[A1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** 21 in the intersection $C \cap D$ **[B1]**

34 in C only and 47 in D only **[B1]**

18 outside both circles **[B1]**

(b) $\frac{21}{120}$ or $\frac{7}{40}$ **[B1]**

(c) $1 - \frac{68}{120}$ or $\frac{34 + 18}{120}$ **[M1]**

$\frac{52}{120}$ or $\frac{13}{30}$ **[A1]**

(d) $\frac{34 + 47}{120}$ or $\frac{120 - 21 - 18}{120}$ **[M1]**

$\frac{81}{120}$ or $\frac{27}{40}$ **[A1]**

[Back to Question] *[Total Marks: 8]*

3 **(a)** $x^2 + 3x + 2$ **[B1]**

(b) $\frac{x^2}{x^2 + 3x + 2} = \frac{8}{15}$ **[M1]**

$15x^2 = 8(x^2 + 3x + 2) \implies 15x^2 = 8x^2 + 24x + 16$ **[M1]**

(c) $(7x + 4)(x - 4) = 0$ or correct substitution into the quadratic formula [M2]
(Award M1 for $(7x + a)(x + b) = 0$ where $ab = -16$ or $7b + a = -24$)

4 **[A1]**

(d) Total counters $= 4^2 + 3(4) + 2 = 30$ and number of blue counters $= 3(4) = 12$ **[M1]**

$\frac{18}{30} = \frac{3}{5}$ **[A1]**

[Back to Question] *[Total Marks: 8]*

4 **(a)** 0.17 **[B1]**

(b) The train does not arrive on time (or equivalent) **[B1]**

[Back to Question] *[Total Marks: 2]*

5 **(a)** $\frac{1}{5} + 0.35 + x + 2x = 1$ or $0.55 + 3x = 1$ **[M1]**

0.15 **[A1]**

(b) $1 - 2(0.15)$ or $\frac{1}{5} + 0.35 + 0.15$ **[M1]**

0.7 or $\frac{7}{10}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

6 **(a)** $(x + 4) + x + (2x - 1) + (3x + 5) = 50$ or $7x + 8 = 50$ **[M1]**

6 **[A1]**

(b) $(2(6) - 1) + (3(6) + 5)$ or $50 - (6 + 4 + 6)$ or $\frac{11 + 23}{50}$ **[M1]**

$\frac{34}{50}$ or $\frac{17}{25}$ or 0.68 **[A1]**

(c) $(6 + 4) + (2(6) - 1)$ or $10 + 11$ or $\frac{21}{50}$ **[M1]**

$\frac{21}{50}$ or 0.42 **[A1]**

[Back to Question] *[Total Marks: 6]*